

PAPER_014b: EMRI Signal Modification by Aether Damping and String Harmonics

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot (1 - U_{bi}/F_U) \cdot e^{-\kappa t}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

Abstract

Extreme Mass Ratio Inspirals (EMRIs) — stellar-mass compact objects spiraling into supermassive black holes — are among the richest GW sources for testing modified gravity. We simulate a UQFF EMRI with $M_{\text{SMBH}} = 106 M_\odot$, $M_{\text{compact}} = 10 M_\odot$ (mass ratio $q = 10.6$) at $D_L = 2.68$ Gpc ($z = 0.5$) over a 2-year LISA observation. Key UQFF predictions: (1) the EMRI signal exhibits 5 string harmonics at frequencies 0.293, 0.586, and 0.879 mHz (the three lowest harmonics of the $f_{\text{ISCO}} = 2.931$ mHz fundamental); (2) the stability factor is enhanced to 1.15 due to U_A Aether damping, which conversely increases EMRI orbital stability; (3) the peak UQFF strain is 5.6548×10^{-23} ; (4) the SNR drops from 100 (GR) to 66.7 (UQFF). Over 1.77×10^5 orbits in 2 years of observation, the accumulated phase lag and string harmonic pattern provide a multi-modal UQFF signature unique to EMRI waveforms.

1. Introduction

EMRIs are among the most information-rich gravitational wave sources because the compact object orbits the SMBH for $\sim 10^4$ – 10^5 cycles within the LISA band, accumulating a precise phase record sensitive to the SMBH spacetime geometry. In GR, the EMRI waveform encodes the Kerr metric to extraordinary precision.

In UQFF, two additional effects modify EMRI waveforms:

1. **Aether damping (U_A):** The Aether compression field couples to the long-duration orbital motion, providing a gradual phase-coherent modulation. For EMRIs at $z = 0.5$, U_A is non-negligible and increases the effective orbital stability.
2. **String harmonics:** The string rotation coupling β_{string} introduces resonant couplings at sub-multiples of the ISCO frequency. For the benchmark system, these appear as 5 harmonic modes with detectable LISA SNR.

The combination of Aether-modified stability and string harmonics creates a UQFF EMRI waveform that is qualitatively distinct from GR predictions.

2. EMRI System Parameters

Parameter	Value
SMBH mass M_{SMBH}	$1.00 \times 10^6 M_{\odot}$
Compact object mass M_c	$10.0 M_{\odot}$
Mass ratio q	1.00×10^{-5}
Redshift z	0.50
Luminosity distance D_L	2.68 Gpc
Observation duration	2.0 yr

3. Orbital Mechanics

3.1 ISCO Frequency

For a Schwarzschild SMBH (non-spinning, as in the benchmark), the ISCO frequency in the source frame is:

$$f_{\text{ISCO,source}} = \frac{c^3}{6^{3/2} \pi G M_{\text{SMBH}}} = f_{\text{ISCO,source}}$$

Redshifted to the observer at $z = 0.5$:

$$f_{\text{ISCO,obs}} = f_{\text{ISCO,source}} / (1 + z) = 2.931 \text{ mHz}$$

This is within LISA's peak sensitivity range (1-10 mHz).

3.2 Orbital Evolution

Over the 2-year observation:

Quantity	Value
f_{ISCO} (observer)	2.931 mHz
Observation duration	2.0 yr
Total EMRI orbits	1.77×10^5
Mean orbital frequency	$\sim 2 \text{ mHz}$ (below ISCO)
Orbital period	$\sim 8.5 \text{ minutes}$

The compact object completes 177,000 full orbits during the observation, each slightly shorter-period as the system evolves toward ISCO.

4. UQFF String Harmonics

4.1 Harmonic Structure

The string coupling β_{string} introduces resonant standing modes in the UQFF vacuum field around the EMRI system. These modes occur at rational fractions of the ISCO frequency (sub-harmonic resonances):

$$f_n = f_{\text{ISCO}} \times (n/N_{\text{harmonics}}), \quad n = 1, 2, \dots, N_{\text{harmonics}}$$

where $N_{\text{harmonics}} = 5$ for the benchmark system.

Harmonic n	Frequency (mHz)	Amplification
1st	0.2931	$\sim \beta_{\text{string}}$ correction

Harmonic n	Frequency (mHz)	Amplification
2nd	0.5862	$\sim \beta_{\text{string}}^2$ correction
3rd	0.8793	$\sim \beta_{\text{string}}^3$ correction
4th	1.1724	—
5th	1.4655 (= $f_{\text{ISCO}} \times 0.5$)	dominant sub-harmonic

The three lowest harmonics (0.293, 0.586, 0.879 mHz) appear as measurable spectral peaks in the EMRI time-frequency representation, detectable via TDI (Time-Delay Interferometry) spectral analysis.

4.2 Physical Origin of Harmonics

The string vacuum field around the EMRI orbit forms a resonant cavity with discrete modes. The orbital motion of the compact object periodically pumps energy into these modes, appearing as sidebands of the EMRI waveform in the frequency domain. This is analogous to parametric resonance but mediated by the string vacuum coupling rather than a mechanical restoring force.

5. Aether Damping and Orbital Stability

5.1 Stability Enhancement

At $z = 0.5$, the Aether field U_A provides a mild restoring effect on the EMRI orbit:

$$\text{Stability factor} = 1 + d_A = 1 + 0.15 = 1.15$$

This 15% stability enhancement means the EMRI orbit is $\sim 15\%$ more stable against run-away inspiral compared to the GR-only prediction. Physically, the Aether buoyancy term provides a slight positive pressure that retards the orbital energy dissipation rate.

Observable consequence: EMRIs in UQFF spend slightly longer (15%) in the LISA band before reaching ISCO, increasing the accumulated phase measurement by the same factor.

5.2 Modified Phase Accumulation

Quantity	GR	UQFF
Orbits in 2 yr	1.77×10^5	$\sim 2.04 \times 10^5$ (stability factor)
Phase accuracy	$s_f \sim 0.001$ rad	$s_f \sim 0.001$ rad
Phase lag vs GR	—	Large (>1000 rad)

6. Strain and SNR Results

6.1 Peak UQFF Strain

The UQFF peak strain for the EMRI at $D_L = 2.68$ Gpc:

$$h_{\text{UQFF,peak}} = 5.6548 \times 10^{-23}$$

This is $\sim 60\times$ smaller than the SMBH merger benchmark ($h_{\text{SMBH}} \sim 4.3 \times 10^{-21}$) due to the small mass ratio $q = 10^{-5}$ reducing the quadrupole emission.

6.2 Signal-to-Noise Ratio

Model	SNR (2-yr coherent)
Standard GR	100
UQFF prediction	66.7
Reduction factor	0.667

The SNR reduction factor 0.667 differs from the $z=0$ value of 0.333 because at $z = 0.5$ the Aether compensation partially offsets the string coupling, yielding $D_{\text{eff}}(z=0.5) \sim 0.667$ for EMRI-type sources.

Both GR and UQFF predictions are above the LISA EMRI detection threshold (SNR ~ 20), ensuring detectability. The factor-of-1.5 SNR difference is measure over a 2-year coherent integration.

7. Multi-Modal UQFF EMRI Signature

The complete UQFF EMRI signature consists of 4 observable components:

Component	Observable	UQFF vs GR
Base waveform	Phase evolution $f(t)$	Phase lag > 1000 rad
SNR	Matched-filter SNR	66.7 (UQFF) vs 100 (GR)
Stability	Time in LISA band	+15% longer (? more cycles)
String harmonics	5 spectral lines at $f_{\text{ISCO}} \times n/5$	Not present in GR

The string harmonic lines at 0.293, 0.586, 0.879 mHz are particularly diagnostic: they appear as narrow spectral features in the LISA TDI data stream at sub-ISCO frequencies, with known frequency ratios (1:2:3). Their absence would rule out string coupling; their presence at the predicted frequencies would confirm UQFF.

8. Comparison: SMBH vs EMRI UQFF Modifications

Property	SMBH Merger ($z=1$)	EMRI ($z=0.5$)
UQFF factor	0.619	~ 0.667
Reduction	38.1%	33.3%
String harmonics	No	Yes (5 modes)
Stability modification	No	+15%
Phase lag	732 GW cycles	> 1000 rad
SNR ratio	0.619	0.667

The EMRI UQFF factor at $z = 0.5$ (0.667) is intermediate between the local value (0.333) and the SMBH value (0.619 at $z=1$), consistent with the smooth redshift evolution of U_A .

9. Testable Predictions

1. **String harmonic lines:** LISA spectral analysis should reveal narrow lines at $f = n \times f_{\text{ISCO}}/5$ for $n = 1-5$ in EMRI signals. Detection probability: $\sim 10\%$ of all EMRIs within LISA horizon.
 2. **Stability factor test:** EMRI in-band lifetimes should be 15% longer than GR predictions, measurable by comparing observed duration to theoretical PN inspiral timescales.
 3. **SNR ratio test:** For EMRIs where independent mass estimates exist, the measured SNR should be $0.667 \times$ the GR-predicted value.
 4. **Phase coherence:** EMRI parameter estimation should find residual phases of > 1000 rad when GR templates are used, pointing toward UQFF-modified templates.
 5. **Rate prediction:** 33.3 EMRI detections/yr (UQFF) vs 50/yr (GR). Three years of LISA operation will provide > 5 s discrimination between these rates.
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10. Conclusions

UQFF modifies EMRI signals in four distinct ways: SNR reduction by factor 0.667 (vs GR), 5 string harmonic spectral lines at sub-ISCO frequencies, 15% stability enhancement from Aether damping, and accumulation of > 1000 rad phase lag over 1.77×10^5 orbits. The predicted LISA EMRI detection rate is 33.3/yr (UQFF) vs 50/yr (GR). The multi-modal nature of UQFF EMRI modifications — involving both waveform amplitude and novel spectral features — makes EMRIs among the best LISA sources for testing the UQFF framework.

References

1. Babak, S. et al., *Science with the space-based interferometer LISA. V: EMRIs*, Phys. Rev. D **95**, 103012 (2017)
 2. Amaro-Seoane, P. et al., *Intermediate and Extreme Mass-Ratio Inspirals in LISA*, Class. Quantum Grav. **24**, R113 (2007)
 3. Barack, L. & Pound, A., *Self-force and radiation reaction in general relativity*, Rep. Prog. Phys. **82**, 016904 (2019)
 4. Murphy, D., `validate_lisa.py` — UQFF LISA EMRI simulation (2026)
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Validator: `validate_lisa.py` — **ALL 3 TESTS PASSED** (TEST 2: EMRI PASS)
 $M_{\text{SMBH}}=106 M_\odot$, $M_{\text{compact}}=10 M_\odot$, $q=10^3$, $z=0.5$, $D_L=2.68$ Gpc;
 $f_{\text{ISCO}}=2.931$ mHz; $\text{orbits}=1.77 \times 10^5$; $\text{observation}=2$ yr;
String harmonics: 5 modes at [0.293, 0.586, 0.879] mHz;
Stability factor=1.15; $h_{\text{UQFF}}=5.6548e-23$; SNR: 100 ? 66.7;
EMRI rate: 50 ? 33.3/yr; ? = 0.0005/day, [SSq] = 0.57

End of Paper 014b

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{SCm} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$Ug_sum + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times Ubi$	Expanding nebulae, stellar winds
Superconductive	$U_{\text{em}} \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.